Dibya Deepta Mishra

Academic Curriculum Vitae

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Research Interests

Industrial organization; structural econometrics; economics of artificial intelligence; causal inference; healthcare markets; economics of education.

Education

Rice University — PhD in Economics, 2024

Fields: Industrial Organization and Structural Econometrics.

Coursework: Advanced Econometrics; Computational Economics; Labor Economics; Industrial Organization I and II; Corporate Finance; Asset Pricing; Financial Econometrics.

Georgia Institute of Technology — MS in Computer Science, 2024

Focus: Machine Learning and Artificial Intelligence.

Coursework: Machine Learning; Machine Learning for Trading; AI Ethics; Natural Language Processing; Algorithms; Stochastic Optimization; Data and Visual Analytics; High-Dimensional Data Analytics.

London School of Economics — MSc in Econometrics and Mathematical Economics, 2017

Indian Institute of Management Calcutta — Post Graduate Diploma in Management, 2014

Indian Institute of Technology Jodhpur — BTech in Electrical Engineering, 2012

Research Appointment

Supervised Program for Alignment Research — Research Fellow

September 2026–present

Study the economic value of AI capability gains using discrete-choice demand models to estimate willingness to pay for capability and safety. Current work includes preliminary estimates and a working draft on model and product design.

Professional Experience

Cornerstone Research — Associate

Chicago, IL | January 2025–September 2026; Summer Associate, 2024

- Estimated structural demand models, demand elasticities, and counterfactual prices; analyzed accounting margins and supported merger reviews with a combined multibillion-dollar transaction value.
- Applied difference-in-differences, synthetic controls, and high-dimensional fixed effects to healthcare and technology datasets with 10M+ observations; developed a theoretical pricing model used in an expert report.
- Assessed reliability and robustness of language and multimodal models in litigation work. Managed 2–3 analysts, reviewed code, and supported reproducible analysis and expert reports.

Amazon Retail Pricing — Economist Intern

Seattle, WA | May–August 2023

Combined product-text embeddings with gradient boosting to forecast demand, reducing RMSE by 5% relative to standard logit demand models. Evaluated discount and bundle prices and presented recommendations to inform future A/B tests.

Publications

AI-assisted teams outperform AI-led teams but not human-only teams in assessing research reproducibility in quantitative social science

With Abel Brodeur et al. Proceedings of the National Academy of Sciences, 123(22), 2026.

Local Population Characteristics and Access Equity of 340B Contract Pharmacies

With Maura Coughlin. Health Affairs Scholar, 3(7), 2025.

The Wizard of OZ (Opportunity Zones): Spatial Spillovers in Place-Based Programs

With Robin C. Sickles and Yanfei Sun. In Seven Decades of Econometrics and Beyond, Springer, 2025, pp. 181–208.

Working Papers and Research Reports

Quality Considerations and Universal Healthcare

Sole author. Working paper, 2026.

Estimates a structural model of hospital choice under limited consideration. Uses neural networks to measure risk-adjusted hospital quality and evaluates information and hospital-participation counterfactuals.

Pennies for Pads: Subsidized Menstrual Health Products and Adoption in India

With Ritika Sethi. Under review, 2026.

Applies doubly robust difference-in-differences to linked survey, administrative, and geospatial data for nearly 500,000 women to study adoption of subsidized menstrual health products.

Beyond a Free Lunch: Unintended Consequences of an Out-of-School Nutrition Program on Adolescent Girls' School Enrollment and Health Outcomes

With Tanmay Devi, Ajinkya Keskar, and Ritika Sethi. Revise and resubmit, Journal of Economic Behavior & Organization, 2026.

Spectrum Allocation to Public Operators: A Structural Analysis of Set-Asides versus Subsidies in Indian Telecom Auctions

With Rajarshi Bhowal and Nurdaulet Menglibayev. Under review, 2026.

Do Language Models Represent Utility-Maximizing Choice?

Sole author. Under review, 2026.

Uses probes and activation-based tests to study decision-relevant representations in language models and their relationship to observed choices.

Detecting Collusion Between AI Agents

Amit Balter, Chengxiao Wang, Dibya Mishra, Fan Wu, and Michael Ball. Apart Research collusion sprint report, October 25, 2026.

Studies behavioral and conversational evidence of collusion in 1,800 agent runs and 18,000 episodes. Contributed to the Python labeling pipeline using Modal. The report evaluates a rule-plus-LLM classifier against 342 annotated episodes and develops measures of collusion intensity, onset, and reciprocity.

Code: github.com/fw2274/detecting-collusion-btw-ai-agents

AI Economics Working Drafts

Safety Screens and AI Product Introduction

Sole author. Working paper, October 2026.

Estimates a recurring-release model using 5,593 developer-weeks for 76 developers; compares timing-based and random safety assessment at equal assessment volume.

Demand for AI Models: Capability, Price, and Safety

With Aryan Rudraraju and Daneal O'Habib. SPAR research draft, October 2026.

Studies associations between price, capability, safety, and OpenRouter usage using 164 models and 7,946 model-week observations.

Research in Progress

Dynamic Competition and Frontier AI Model Release

AI Agent Swarms and Collective Risk

Who Matches with Whom: Mergers and Acquisitions in Health Systems

Allocating Bundles: An Operations Research Framework — with Jeremy Fox

Long Distance Relationships: Analyzing Hospital-Pharmacy Contracts — with Maura Coughlin

Earlier Research and Professional Experience

Indian School of Business — Research Associate

Hyderabad, India | September 2017–April 2019

Research with Tarun Jain and Sisir Debnath. Estimated healthcare consumer-choice and firm-entry models; evaluated market and policy interventions using administrative and survey data, difference-in-differences, instrumental variables, and matching.

Rice University — Research Assistant: Jeremy Fox (2021; 2022–2023); Mallesh Pai (2021–2022); Maura Coughlin (2022).

City University London — Research Assistant for Fabrice Defever, 2017.

Age UK, London — Research Assistant for Jose Iparraguirre, 2015–2016.

HSBC, Mumbai — Business and Regulatory Risk Research, May 2014–March 2015. Assessed regulatory and macroeconomic developments and presented risk findings to senior management.

Teaching Experience

Rice University — Graduate Teaching Assistant

2020–2024

PhD Microeconomics II; PhD Econometrics I; Math Camp; Industrial Organization; Macroeconomics II.

London School of Economics — Graduate Teaching Assistant

2016–2017 | Intermediate Microeconomics

Indian School of Business — Teaching Assistant

2017–2018 | Managerial Economics; Healthcare Economics; Economics of Healthcare Markets

Honors Fellowships and Grants

Social Sciences Research Institute Dissertation Research Improvement Grant
Rice University, 2024

George R. Zodrow Award for Best Performance in a Workshop
Rice University, 2024

Royal Economic Society Conference Grant
2023

Social Sciences Travel Research Grants
Rice University, 2021–2022 and 2022–2023

W. Michael Carter Fellowship in Economics
Rice University, 2021

Rica and Tony Ligeralde Award for Best Performance in the Theory Qualifying Exam
Rice University, 2021

Dagobert L. Brito Award for Best Performance by a Recitation Instructor in Economics
Rice University, 2021

Maria Esther and Carlos Linares Award for Best Performance in Economics First-Year Courses
Rice University, 2020

Graduate Fellowship
Rice University, 2019–2024

Inlaks Scholar
London School of Economics, 2015–2017. One of 12 students across India awarded a full scholarship for master's study.

Charpak Scholar
French Embassy, 2013. Full scholarship for exchange at EMLYON.

Viterbi India Fellow
Indo-US Science and Technology Forum, 2011. One of 10 students selected across India for research at the University of Southern California.

National Talent Search Examination Scholar
Government of India, 2006. One of 750 scholarship recipients across India.

Conferences

2024

Texas Camp Econometrics; Development and Political Economics Conference (DevPEC); ASHEcon; Development Economics Workshop, University of Texas at Austin; North American Summer Meeting of the Econometric Society.

2023

RES and SES Annual Conference; ASHEcon; SEA Annual Conference; Association for Public Policy Analysis and Management Annual Conference; Annual Conference of Economic Development and Growth; DSE Winter School; Econometric Society Asia Meeting.

2022

ASHEcon.

Professional Service

Referee: PLOS ONE; Economic Modelling.

Workshops and Advanced Training

Optimization-Conscious Econometrics Summer School, University of Chicago — June 2023

Machine Learning and Heterogeneous Causal Effects — May 2023

RES Easter Training School, University of Bristol — March 2023

Causal Inference — February 2023

DSE Structural Econometrics Bootcamp, online — December 2022

DSE Market Design Bootcamp, MIT — August 2022

NBER Health Economics Bootcamp, University of California, Berkeley — June 2022

DSE Structural Econometrics Bootcamp, online — August 2021

Structural Methods in Finance — June 2021

Partial Identification in Practice — May 2021

Math-Econ-Code — January and June 2021

AEA Continuing Education in Industrial Organization, online — January 2021

Structural Methods in Accounting — August 2020

Professional Publications and Public Writing

Incentives in the Healthcare System

With Sisir Debnath and Tarun Jain. Harvard Business Publishing case study, 2019.

Panel Summary: What's next for DOJ and FTC?

With Julia Lagnese, Arshia Hashemi, and Haoran Zhang. American Bar Association, Antitrust Law Section.

You Can't Reward What You Can't See

Alignment × Economics blog series, September 2026.

Applies Holmström's informativeness principle to AI monitoring, process supervision, and reward hacking.

Research Tools

Pennies for Pads Results and Siting Planner

Interactive research companion for exploring efficiency and equity tradeoffs in subsidized pharmacy locations.

Methods and Technical Skills

Econometrics: Structural discrete-choice and demand models; causal inference; difference-in-differences; synthetic controls; instrumental variables; matching; high-dimensional fixed effects; double machine learning; doubly robust estimation; counterfactual simulation.

Machine learning and experimentation: Gradient boosting; neural networks; product-text embeddings; language-model representation analysis; model validation; A/B test analysis; prediction-assisted inference; Monte Carlo simulation.

Programming and data: Python (pandas, NumPy, scikit-learn, PyTorch); SQL; R; Stata; Julia; MATLAB; Octave; Git.

Cloud and research software: AWS EC2 and SageMaker; Apache Spark; Modal; QGIS; ArcGIS; LaTeX; Microsoft Office.

Professional Credentials

Financial Risk Manager (FRM), 2020. Passed Levels I and II of the CFA Program.